

Zizhe Zhang

email: zizhe@cs.jhu.edu
[website](#) | [github](#) | [linkedin](#) | [scholar](#)

EDUCATION

Johns Hopkins University , Whiting School of Engineering Doctor of Philosophy (Ph.D.), Computer Science Advisor: Prof. Haimin Hu	Baltimore, MD Aug 2026 – May 2030 (expected)
University of Pennsylvania , GRASP Lab Master of Science in Engineering (M.S.E.), Robotics Advisor: Prof. Nadia Figueroa	Philadelphia, PA Aug 2024 – May 2026
University of California San Diego , Jacobs School of Engineering Visiting Student	San Diego, CA Jan 2023 – Mar 2023
Southeast University , School of Instrument Science and Engineering Bachelor of Engineering (B.E.) in Measuring Control Technology & Instruments Advisor: Prof. Yuan Yang	Nanjing, China Aug 2020 – Jun 2024

HONORS AND AWARDS

Best Contribution Award , IROS Workshop on Exploring the Role of Energy in Robot Learning and Control	2025
Best Student Paper Award , IROS Workshop on Building Safe Robots	2025
GAPSA Individual Grant , University of Pennsylvania	2025
IROS-SDC Travel Award , IROS 2025	2025
Outstanding Bachelor's Thesis , Southeast University	2024
Merit Student Award , Southeast University	2021

PUBLICATIONS

(* and † denote equal contribution)

Peer-Reviewed Conference Articles

C3. Viability-Preserving Passive Torque Control

Zizhe Zhang*, Yicong Wang*, Zhiquan Zhang, Tianyu Li, and Nadia Figueroa
IEEE International Conference on Robotics & Automation (ICRA), 2026 [[arXiv](#)] [[code](#)] [[website](#)]
IROS Workshop on Building Safe Robots (**Best Student Paper Award**), 2025
IROS Workshop on Exploring the Role of Energy in Robot Learning and Control (**Best Contribution Award**), 2025

C2. Flow with the Force Field: Learning 3D Compliant Flow Matching Policies from Force and Demonstration-Guided Simulation Data

Tianyu Li*, Yihan Li*, **Zizhe Zhang**, and Nadia Figueroa
IEEE International Conference on Robotics & Automation (ICRA), 2026 [[arXiv](#)] [[website](#)]
IROS Workshop on Exploring the Role of Energy in Robot Learning and Control, 2025
IROS Workshop on Contact and Impact-aware Manipulation, 2025

C1. VLMgineer: Vision Language Models as Robotic Toolsmiths

George Jiayuan Gao*, Tianyu Li*, Junyao Shi, **Zizhe Zhang**[†], Yihan Li[†], Nadia Figueroa, and Dinesh Jayaraman
International Conference on Learning Representations (ICLR), 2026 [[arXiv](#)] [[Code](#)] [[website](#)]
Robotics: Science and Systems (RSS) Workshop on Robot Hardware-Aware Intelligence (**Oral Spotlight**), 2025

Peer-Reviewed Journal Articles

J1. Image-Based Visual Servoing for Enhanced Cooperation of Dual-Arm Manipulation

Zizhe Zhang, Yuan Yang, Wenqiang Zuo, Guangming Song, Aiguo Song, and Yang Shi
IEEE Robotics and Automation Letters (RA-L), 2025 [[arXiv](#)] [[website](#)]

RESEARCH EXPERIENCE

University of Pennsylvania Research Engineer Graduate Research Assistant <u>Figueroa Robotics Lab</u> , advised by Prof. Nadia Figueroa	Philadelphia, PA Jun 2026 - Aug 2026 Nov 2024 – May 2026
Duke University Visiting Scholar <u>Robot Dexterity Lab</u> , hosted by Prof. Xianyi Cheng	Durham, NC Jun 2025 – Aug 2025
Southeast University Undergraduate Research Assistant <u>Robotic Perception and Control Lab</u> , advised by Prof. Yuan Yang	Nanjing, China Dec 2023 – Aug 2024
Undergraduate Research Assistant <u>Automotive Safety Technology Lab</u> , advised by Prof. Dong Wang	Mar 2022 – Jun 2023
Undergraduate Research Assistant <u>Advanced Navigation Technology Lab</u> , advised by Prof. Xuanpeng Li	Jul 2021 – Sep 2021

TEACHING EXPERIENCE

University of Pennsylvania Teaching Assistant ESE 5000 Linear Systems Theory	Philadelphia, PA Fall 2025 Instructor: Prof. George Pappas
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SERVICE

Program Committee NeurIPS Workshop on Physical World AI	2026
Reviewer IEEE Robotics and Automation Letters (RA-L) IEEE Transactions on Robotics (TRO) IEEE Transactions on Industrial Electronics (TIE) IEEE Transactions on Industrial Informatics (TII) IEEE/CAA Journal of Automatica Sinica (JAS) Machine Intelligence Research (MIR) IEEE International Conference on Robotics and Automation (ICRA)	2026
Volunteer Robotics: Science and Systems (RSS)	2025

Professional Memberships

Robot Learning Foundation Member	2026 - Present
IEEE Robotics and Automation Society Member	2024 - Present
IEEE Young Professionals Member	2024 - Present
IEEE Member	2024 - Present

SKILLS

Programming	C, C++, Python, MATLAB
Tools	CMake, PyTorch, ROS, RLBench, ManiSkill, CoppeliaSim, PyBullet, Isaac Gym/Sim/Lab, MuJoCo
Robots	UR3, UR3e, Franka Emika Panda, Franka Research 3